Faculty
Civil Engineering and GeoSciences

Partial Exam	**CT1000S**
	Structural Mechanics, Stability
Total number of pages	**8 pages (excl cover)**
Date and time	**31 JAN 2025** from **09:00-12:00**
Responsible lecturer	**J.W. Welleman**

Only the work / answers written on examination paper will be assessed, unless otherwise specified under 'Additional Information'.

Exam questions (to be filled in by course examiner)

Total number of questions: 2

☒ **questions may differ in weight** *(the time mentioned is an indicator for the weight)*

Use of tools and sources of information (to be filled in by course examiner)

Not allowed:

- Mobile phone, smart Phone or devices with similar functions.
- Answers written with red pen or with pencils.
- Calculators with CAS and/or wifi/BT and/or PDF capabilities
- Tools and/or sources of information unless otherwise specified below.

Allowed:

☐ **books** ☐ **notes** ☐ **dictionaries** ☐ **syllabus**

☐ **formula sheets (see also below under 'additional information')** ☒ **calculators**

☐ **computer** ☐ **...**

☒ **scientific (graphical)calculator** ☒ **drawing material**

Additional information (if necessary to be filled in by the examiner)

- **Use for each problem a separate examination paper**
- **The question form contains fomula sheets which can be used.**
- **Students can take the question form home after the exam.**
- **Specify the correct BSc or MSc course code on the exam paper**
- **BSc students are allowed to answer in Dutch**
- **No student leaves without delivering an exam paper with a name on it!**

Exam graded by: (the marking period is 15 working days at most)

Every suspicion of fraud is reported to the Board of Examiners.

Mobile Phone OFF.

Problem 1 : Theory (approx. 20 min)

A stability assessment is required for the structure presented in fig. 1. The connections in both C and D are hinged and r is a linear elastic rotational spring. Only column AC can bend, all other members are rigid. Any axial deformation is assumed to be negligible.

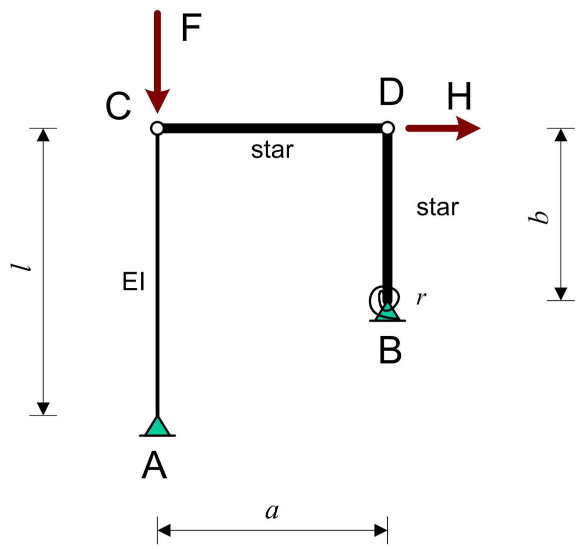


Figure 1 : Structure with concentrated forces

NOTE:
Modelling involves a sketch of the model, a description (formula) of possible springs used and or boundary conditions needed.

Questions:

a) Sketch the expected mode(s) for the limit of a stable equilibrium.
b) Simplify this structural model to a mathematical model for which the limit load can be found.
c) Explain the steps involved to find the limit load. You do not need to solve for the expression of the answer.

Problem 2 : Buckling and 2nd order effect (approx. 40 min)

A frame structure is loaded with two concentrated vertical forces F as is shown in fig. 2. Apart from this vertical load also a horizontal load H is applied. Bending stiffnesses expressed in EI can be depicted from fig. 2 and worth to mention that the upper and lower beams in the frame have different bending stiffnesses. The length l of the structure is 10 m, the height h is 5 m. The influence of any axial deformation can be neglected in the problem and all connections between columns and beams are assumed to be rigid.

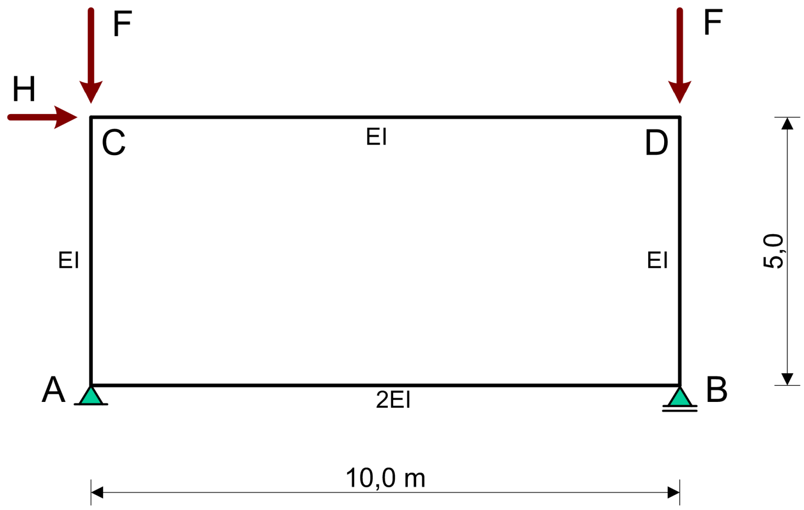


Figure 2 : Frame loaded with concentrated forces

Given: $EI = 15000$ kNm^2; $F = 1000$ kN ; $H = 90$ kN;

From a *geometrical linear analysis* is known that the maximum moment in column AC is given as:

$$M_{\max}^{AC} = \frac{(l+3h)}{3(l+4h)} Hh$$

Also from this analysis is known that the first order horizontal displacement at C can be expressed as:

$$u_{C-1} = \frac{(l^2+6hl+6h^2)}{36(l+4h)} \frac{Hh^2}{EI}; \qquad \text{(check this at home if you like ...)}$$

Questions:

a) Sketch the buckling mode and explain how to find the buckling load for this problem.
b) Find the buckling load in [kN] for the given situation and support your approach if needed with sketches.
c) Find the first order moment in A and C in [kNm].
d) Find the second order displacement in [m] of point C for the given F and H
e) Find the second order moment in A in [kNm] for the given F and H.

FORMULAS

Eulerian Buckling:

$$F_k = \frac{\pi^2 EI}{l_k^2}$$

$$\frac{1}{F_\text{k}} = \frac{1}{\frac{r}{l}} + \frac{1}{\frac{\pi^2 EI}{4l^2}} \quad \Rightarrow \quad l_\text{k} = l\sqrt{4 + \frac{10}{\rho}} \quad \text{met}: \rho = \frac{rl}{EI}$$

Kinematic and Constitutive relations:

$$\varphi = -\frac{dw}{dx} \quad \kappa = \frac{d\varphi}{dx} \quad M = EI\kappa$$

Differential equation:

$$w'''' + \alpha^2 w'' = 0 \quad \text{with: } \alpha^2 = \frac{F}{EI}$$

$$w(x) = C_1 + C_2 x + C_3 \cos\alpha x + C_4 \sin\alpha x$$

$$S_z(x) = M' - Fw' \quad \text{with:} \quad S_z(x) = -F.C_2$$

$$F_\text{k} = \frac{(\eta_1 + \eta_2)^2}{\eta_1\eta_2(\eta_1 + \eta_2 - 4)} \times \frac{\pi^2 EI}{l^2} \quad \text{with: } \begin{array}{l} \eta_1 = 4 + \frac{10}{\rho_1}\,;\ \rho_1 = \frac{r_1 l}{EI} \\ \eta_2 = 4 + \frac{10}{\rho_2}\,;\ \rho_2 = \frac{r_2 l}{EI} \end{array}$$

$$F_k = \frac{(5 + 2\rho_1)(5 + 2\rho_2)}{(5 + \rho_1)(5 + \rho_2)} . \frac{\pi^2 EI}{l^2} \quad \text{with:} \quad \rho_1 = \frac{r_1 l}{EI} \quad \rho_2 = \frac{r_2 l}{EI}$$

Gonio results:

$$\tan\alpha l = \alpha l \quad \Rightarrow \quad \alpha l \cong 4.49 \cong \frac{\pi}{0.7}$$

Deformation energy:

$$E_v = \int \tfrac{1}{2} EA\varepsilon^2 \mathrm{d}x \quad \text{(extension)}$$

$$E_v = \int \tfrac{1}{2} EI\kappa^2 \mathrm{d}x \quad \text{(bending)}$$

Kinematic relations:

$$\varepsilon = \frac{\mathrm{d}u}{\mathrm{d}x}$$

$$\kappa = -\frac{\mathrm{d}^2 w}{\mathrm{d}x^2}$$

Complementary energy:

$$E_c = \int \frac{N^2}{2EA} \mathrm{d}x \quad \text{(extension)}$$

$$E_c = \int \frac{M^2}{2EI} \mathrm{d}x \quad \text{(bending)}$$

Constitutive relations:

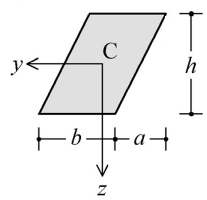
$$N = EA.\varepsilon$$

$$M = EI.\kappa$$

Rayleigh:

$$F_k = \frac{E_v}{\int \frac{1}{2}\left(\frac{\mathrm{d}w}{\mathrm{d}x}\right)^2 \mathrm{d}x}$$

Math tools:



$$I_{yy} = \tfrac{1}{12}(a^2 + b^2)bh$$

$$I_{zz} = \tfrac{1}{12}bh^3$$

$$I_{yz} = \tfrac{1}{12}abh^2$$

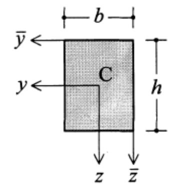
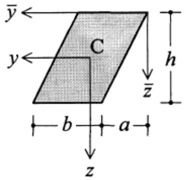
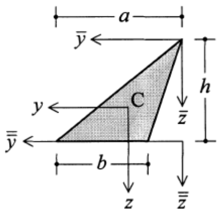
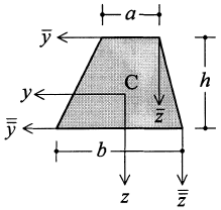
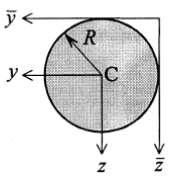
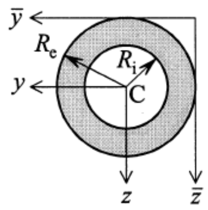
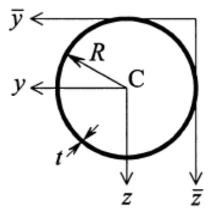
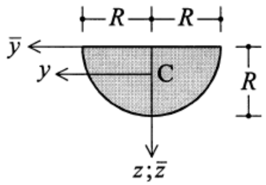
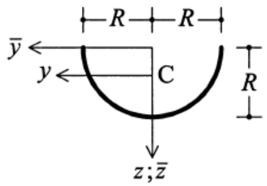
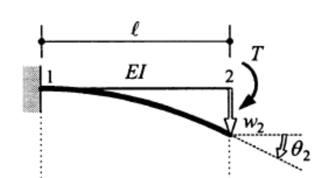
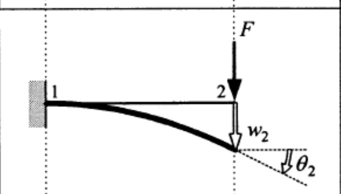
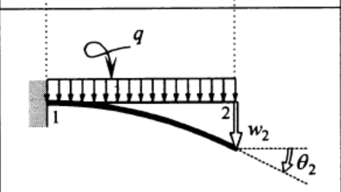
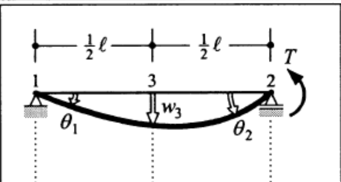
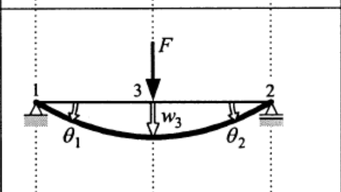
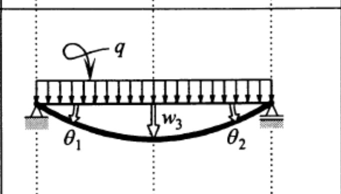
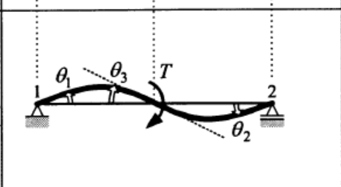
Figure	Area, coordinates centroid C	Second moments of area	
		centroidal	other
	Rectangle $A = bh$ $\bar{y}_C = \frac{1}{2}b$ $\bar{z}_C = \frac{1}{2}h$	$I_{yy} = \frac{1}{12}b^3h$ $I_{zz} = \frac{1}{12}bh^3$ $I_{yz} = 0$	$I_{\bar{y}\bar{y}} = \frac{1}{3}b^3h$ $I_{\bar{z}\bar{z}} = \frac{1}{3}bh^3$ $I_{\bar{y}\bar{z}} = \frac{1}{4}b^2h^2$
	Parallelogram $A = bh$ $\bar{y}_C = \frac{1}{2}(a+b)$ $\bar{z}_C = \frac{1}{2}h$	$I_{yy} = \frac{1}{12}(a^2+b^2)bh$ $I_{zz} = \frac{1}{12}bh^3$ $I_{yz} = \frac{1}{12}abh^2$	$I_{\bar{z}\bar{z}} = \frac{1}{3}bh^3$
	Triangle $A = \frac{1}{2}bh$ $\bar{y}_C = \frac{1}{3}(2a-b)$ $\bar{z}_C = \frac{2}{3}h$	$I_{yy} = \frac{1}{36}(a^2-ab+b^2)bh$ $I_{zz} = \frac{1}{36}bh^3$ $I_{yz} = \frac{1}{72}(2a-b)bh^2$	$I_{\bar{z}\bar{z}} = \frac{1}{4}bh^3$ $I_{\bar{y}\bar{z}} = \frac{1}{8}(2a-b)bh^2$ $I_{\bar{\bar{z}}\bar{\bar{z}}} = \frac{1}{12}bh^3$
	Trapezium $A = \frac{1}{2}(a+b)h$ $\bar{z}_C = \frac{1}{3}\frac{a+2b}{a+b}h$	$I_{zz} = \frac{1}{36}\frac{a^2+4ab+b^2}{a+b}h^3$	$I_{\bar{z}\bar{z}} = \frac{1}{12}(a+3b)h^3$ $I_{\bar{\bar{z}}\bar{\bar{z}}} = \frac{1}{12}(3a+b)h^3$
	Circle $A = \pi R^2$	$I_{yy} = I_{zz} = \frac{1}{4}\pi R^4$ $I_{yz} = 0$ $I_p = \frac{1}{2}\pi R^4$	$I_{\bar{y}\bar{y}} = I_{\bar{z}\bar{z}} = \frac{5}{4}\pi R^4$ $I_{\bar{y}\bar{z}} = \pi R^4$

Figure	Area, coordinates centroid C	Second moments of area	
		centroidal	other
	Thick-walled ring $A = \pi(R_e^2 - R_i^2)$	$I_{yy} = I_{zz} = \frac{1}{4}\pi(R_e^4 - R_i^4)$ $I_{yz} = 0$ $I_p = \frac{1}{2}\pi(R_e^4 - R_i^4)$	
	Thin-walled ring $A = 2\pi Rt$	$I_{yy} = I_{zz} = \pi R^3 t$ $I_{yz} = 0$ $I_p = 2\pi R^3 t$	$I_{\bar{y}\bar{y}} = I_{\bar{z}\bar{z}} = 3\pi R^3 t$
	Semicircle $A = \frac{1}{2}\pi R^2$ $\bar{y}_C = 0$ $\bar{z}_C = \frac{4}{3\pi}R$ $= 0.424R$	$I_{yy} = \frac{1}{8}\pi R^4 = 0.393R^4$ $I_{zz} = (\frac{\pi}{8} - \frac{8}{9\pi})R^4$ $= 0.110R^4$ $I_{yz} = 0$	$I_{\bar{y}\bar{y}} = I_{\bar{z}\bar{z}} = \frac{1}{8}\pi R^4$ $I_{\bar{y}\bar{z}} = 0$
	Semicircular ring $A = \pi Rt$ $\bar{y}_C = 0$ $\bar{z}_C = \frac{2}{\pi}R$ $= 0.637R$	$I_{yy} = \frac{1}{2}\pi R^3 t$ $I_{zz} = (\frac{\pi}{2} - \frac{4}{\pi})R^3 t$ $= 0.298R^3 t$ $I_{yz} = 0$	$I_{\bar{y}\bar{y}} = I_{\bar{z}\bar{z}} = \frac{1}{2}\pi R^3 t$ $I_{\bar{y}\bar{z}} = 0$

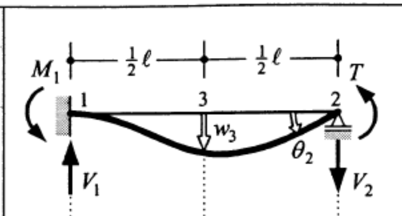
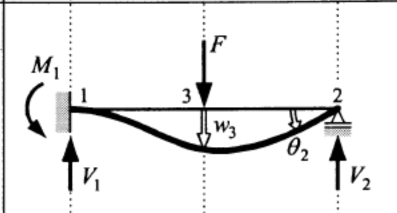
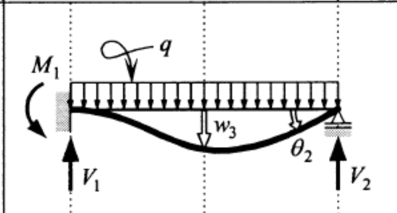
forget-me-nots

(1)		$\theta_2 = \frac{T\ell}{EI};\quad w_2 = \frac{T\ell^2}{2EI}$
(2)		$\theta_2 = \frac{F\ell^2}{2EI};\quad w_2 = \frac{F\ell^3}{3EI}$
(3)		$\theta_2 = \frac{q\ell^3}{6EI};\quad w_2 = \frac{q\ell^4}{8EI}$

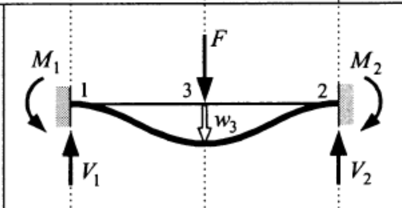
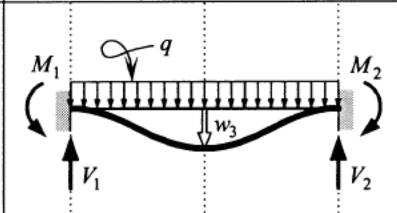
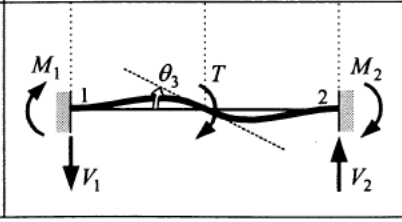
simply supported beam (statically determinate)

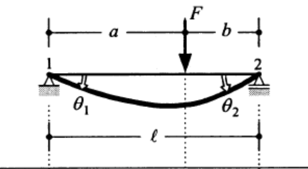
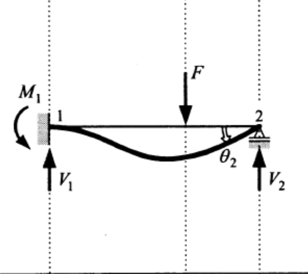
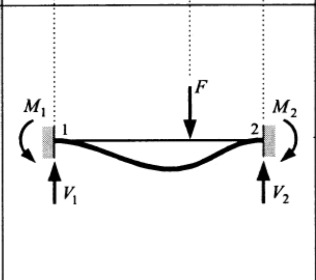
(4)		$\theta_1 = \frac{1}{6}\frac{T\ell}{EI};\quad \theta_2 = \frac{1}{3}\frac{T\ell}{EI};\quad w_3 = \frac{1}{16}\frac{T\ell^2}{EI}$
(5)		$\theta_1 = \theta_2 = \frac{1}{16}\frac{F\ell^2}{EI};\quad w_3 = \frac{1}{48}\frac{F\ell^3}{EI}$
(6)		$\theta_1 = \theta_2 = \frac{1}{24}\frac{q\ell^3}{EI};\quad w_3 = \frac{5}{384}\frac{q\ell^4}{EI}$
(a)		$\theta_1 = \theta_2 = \frac{1}{24}\frac{T\ell}{EI};\quad \theta_3 = \frac{1}{12}\frac{T\ell}{EI};\quad w_3 = 0$

statically indeterminate beam (one fixed end)

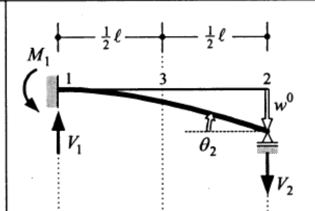
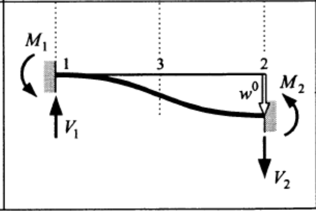
(7)		$\theta_2 = \frac{1}{4}\frac{T\ell}{EI};\quad w_3 = \frac{1}{32}\frac{T\ell^2}{EI}$ $M_1 = \frac{1}{2}T;\quad V_1 = V_2 = \frac{3}{2}\frac{T}{\ell}$
(8)		$\theta_2 = \frac{1}{32}\frac{F\ell^2}{EI};\quad w_3 = \frac{7}{768}\frac{F\ell^3}{EI}$ $M_1 = \frac{3}{16}F\ell;\quad V_1 = \frac{11}{16}F;\quad V_2 = \frac{5}{16}F$
(9)		$\theta_2 = \frac{1}{48}\frac{q\ell^3}{EI};\quad w_3 = \frac{1}{192}\frac{q\ell^4}{EI}$ $M_1 = \frac{1}{8}q\ell^2;\quad V_1 = \frac{5}{8}q\ell;\quad V_2 = \frac{3}{8}q\ell$

statically indeterminate beam (two fixed ends)

(10)		$w_3 = \frac{1}{192}\frac{F\ell^3}{EI}$ $M_1 = M_2 = \frac{1}{8}F\ell;\quad V_1 = V_2 = \frac{1}{2}F$
(11)		$w_3 = \frac{1}{384}\frac{q\ell^4}{EI}$ $M_1 = M_2 = \frac{1}{12}q\ell^2;\quad V_1 = V_2 = \frac{1}{2}q\ell$
(b)		$\theta_3 = \frac{1}{16}\frac{T\ell}{EI};\quad w_3 = 0$ $M_1 = M_2 = \frac{1}{4}T;\quad V_1 = V_2 = \frac{3}{2}\frac{T}{\ell}$

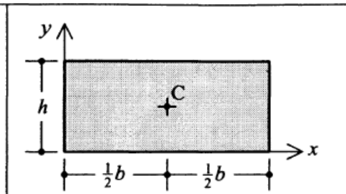
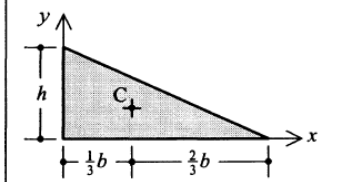
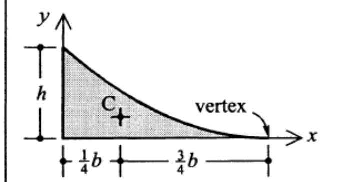
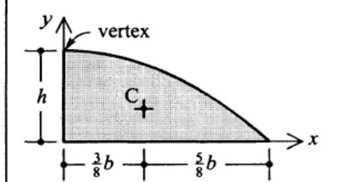
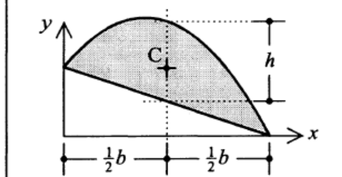
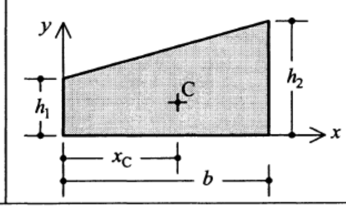
(c)		$\theta_1 = \frac{Fab(\ell+b)}{6EI\ell} = \frac{F\ell^2}{6EI}\left(2\frac{a}{\ell} - 3\frac{a^2}{\ell^2} + \frac{a^3}{\ell^3}\right)$ $\theta_2 = \frac{Fab(\ell+a)}{6EI\ell} = \frac{F\ell^2}{6EI}\left(\frac{a}{\ell} - \frac{a^3}{\ell^3}\right)$
(d)		$M_1 = \frac{Fb(\ell^2-b^2)}{2\ell^2} = F\ell\left(\frac{a}{\ell} - \frac{3}{2}\frac{a^2}{\ell^2} + \frac{1}{2}\frac{a^3}{\ell^3}\right)$ $V_1 = \frac{Fb(3\ell^2-b^2)}{2\ell^3} = F\left(1 - \frac{3}{2}\frac{a^2}{\ell^2} + \frac{1}{2}\frac{a^3}{\ell^3}\right)$ $V_2 = \frac{Fa^2(3\ell-a)}{2\ell^3} = F\left(\frac{3}{2}\frac{a^2}{\ell^2} - \frac{1}{2}\frac{a^3}{\ell^3}\right)$ $\theta_2 = \frac{Fa^2b}{4EI\ell} = \frac{F\ell^2}{4EI}\left(\frac{a^2}{\ell^2} - \frac{a^3}{\ell^3}\right)$
(e)		$M_1 = \frac{Fab^2}{\ell^2} = F\ell\left(\frac{a}{\ell} - 2\frac{a^2}{\ell^2} + \frac{a^3}{\ell^3}\right)$ $V_1 = \frac{Fb^2(\ell+2a)}{\ell^3} = F\left(1 - 3\frac{a^2}{\ell^2} + 2\frac{a^3}{\ell^3}\right)$ $M_2 = \frac{Fa^2b}{\ell^2} = F\ell\left(\frac{a^2}{\ell^2} - \frac{a^3}{\ell^3}\right)$ $V_2 = \frac{Fa^2(\ell+2b)}{\ell^3} = F\ell\left(3\frac{a^2}{\ell^2} - 2\frac{a^3}{\ell^3}\right)$

support reactions and rotations at the beam ends

(f)		$M_1 = \frac{3EI}{\ell^2}w^0;\quad V_1 = V_2 = \frac{3EI}{\ell^3}w^0$ $\theta_2 = \frac{3}{2}\frac{w^0}{\ell}$ $\theta_3 = \frac{9}{8}\frac{w^0}{\ell};\quad w_3 = \frac{5}{16}w^0$
(g)		$M_1 = M_2 = \frac{6EI}{\ell^2}w^0;\quad V_1 = V_2 = \frac{12EI}{\ell^3}w^0$ $\theta_3 = \frac{3}{2}\frac{w^0}{\ell};\quad w_3 = \frac{1}{2}w^0$

settlements

properties of plane figures to be used for the moment-area theorems

(1)		rectangle: $y = h$ $A = bh$ $x_C = \frac{1}{2}b$
(2)		triangle: $y = h\left\{1 - \frac{x}{b}\right\}$ $A = \frac{1}{2}bh$ $x_C = \frac{1}{3}b$
(3)		parabola: $y = h\left\{1 - \frac{x}{b}\right\}^2$ $A = \frac{1}{3}bh$ $x_C = \frac{1}{4}b$
(4)		parabola: $y = h\left\{1 - \left(\frac{x}{b}\right)^2\right\}$ $A = \frac{2}{3}bh$ $x_C = \frac{3}{8}b$
(5)		parabola: $A = \frac{2}{3}bh$ $x_C = \frac{1}{2}b$
(6)		trapezium: $y = h_1 + (h_2 - h_1)\frac{x}{b}$ $A = \frac{1}{2}b(h_1 + h_2)$ $x_C = \frac{1}{3}b\frac{h_1 + 2h_2}{h_1 + h_2}$

ANSWERS

Problem 1

a) Sketch two possible shapes, one in which the column does not bend and one in which it bends. (Both cases can be with sway but this is not part of the grading.)

b) The horizontal load is not relevant for finding the critical or buckling load.

Put the structure in a displaced mode, for that use a horizontal displacement u at C. It then becomes clear that the supporting structure with rigid members ad rotational spring can be substituted by a horizontal equivalent spring at C. In this model the equivalent spring stiffness becomes:

$$k = \frac{r}{b^2} \quad \text{[kN/m]}$$

c) Since the compressive member is not rigid, an Euler model is used in which point C is supported with a horizontal spring and at B a simple support.

Equilibrium in the deformed state is given with the governing ODE:

$$EI\frac{\mathrm{d}^4 w(x)}{\mathrm{d}x^4} + F\frac{\mathrm{d}^2 w(x)}{\mathrm{d}x^2} = 0$$

General solution:

$$w(x) = C_1 + C_2 x + C_3 \cos\alpha x + C_4 \sin\alpha x$$

with: $\alpha^2 = \dfrac{F}{EI}; \quad S_z = -FC_2 = -EI\alpha^2 C_2$

This model requires four boundary conditions:

$$x = 0; \quad M = 0; \quad S_z = kw;$$
$$x = l; \quad M = 0; \quad w = 0;$$

This problem has 5 unknowns. Substituting the general solution into the boundary conditions yields to four homogeneous equations. A non-trivial solution can be found only if the determinant of the matrix which belongs to the system of four homogeneous equations is zero. The lowest value of the root of this characteristic function will result into the lowest limit value for which the problem is in stable equilibrium.

Not asked for:

Characteristic function (determinant=0) result in:

$$\left(-EI\alpha^2 + kl\right)EI\alpha^4 \sin\alpha l = 0$$

$$EI\alpha^2 = kl \quad or \quad \sin\alpha l = 0$$

$$F_{k-1} = kl \quad or \quad F_{k-1} = \frac{\pi^2 EI}{l^2}$$

Problem 2

a) Door H op nul te stellen ontstaat een knikprobleem. De kolom is een ongeschoorde aan twee zijden verend ingeklemde staaf. Hiervoor kan de eta-formule worden gebruikt, correcte schets vereist waaruit blijkt dat de kniklengte groter is dan 5 m en waaruit de verende werking van de beide regels blijkt (met de minste stijfheid).

b) De verende werking van de boven- en onderregel kan worden bepaald:

$$r_1 = r_{onderregel} = \frac{6\times 2EI}{10}; \quad \eta_1 = \frac{r_1\times 5}{EI} = \frac{17}{3}$$
$$r_2 = r_{bovenregel} = \frac{6\times EI}{10}; \quad \eta_2 = \frac{r_2\times 5}{EI} = \frac{22}{3}$$
$$F_k = \frac{(\eta_1+\eta_2)^2}{(\eta_1\eta_2(\eta_1+\eta_2-4))}\times\frac{\pi^2 EI}{5^2} = \frac{50700\pi^2}{187} = 2675 \text{ kN}$$

c) Grootste moment ten gevolge van alleen H treedt op in A of B en kan worden bepaald met de gegeven formule. Dit moment bedraagt 125 kNm. Het moment in C en D bedraagt 100 kNm. Dit kan worden gevonden op basis van evenwicht met de gegeven uitdrukking voor het moment in A of door een “nette” berekening met behulp van de hybride krachtenmethode (werd niet gevraagd).

d) De 2[e] orde verplaatsing in C wordt gevonden met behulp van de vegrotingsfactor.

$$\frac{n}{n-1} = 1{,}5967 \quad \text{met: } n = \frac{2675}{1000} = 2{,}675$$
$$u_1 = 0{,}0763 \text{ m}$$
$$u_2 = 1{,}5967\times u_1 = 1{,}6\times 0{,}0763 = 0{,}1218 \text{ m}$$

e) Een globale afschatting van het 2[e] orde moment volgt uit het 1[e] orde moment door toepassen van de vergrotingsfactor:

$$M_1 = 125 \text{ kNm}$$
$$u_2 = 1{,}5967\times M_1 = 199{,}6 \text{ kNm}$$

Een betere benadering wordt gevonden door uit te gaan van (buigpunt kolom ongeveer halverwege vanwege kop en voetmoment van resp. 100 en 125 kNm):

$$M_2 = M_1 + F\times\tfrac{1}{2}u_2 = M_1 + \tfrac{1}{2}\times F\times\frac{n}{n-1}u_1 = 186 \text{ kNm}$$
$$\text{met: } u_1 = 0{,}0763 \text{ m}$$

Beide antwoorden worden goed gerekend.